

Rafi Mahmud Tumon

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Education

Technical University Berlin	Berlin, Germany
<i>Master of Science in Geodesy and Geoinformation Science</i>	<i>October 2020 – Present</i>
• Specialization: Geoinformation Science; Key Coursework: Photogrammetric Computer Vision, Geoinformatics, Geodatabases and Infrastructures, Space Geodesy, Adjustment Theory, Deep Learning for Geographic Data	
Rajshahi University of Engineering and Technology	Rajshahi, Bangladesh
<i>Bachelor of Science in Urban and Regional Planning</i>	<i>March 2014 – November 2018</i>

Work Experience

German Aerospace Center (DLR)	Berlin, Germany
<i>Student Assistant</i>	<i>January 2026 – Present</i>
• Application of machine learning methods with multimodal geodata in urban areas. • Utilize Python and QGIS for geospatial analysis and model development.	
OXG Glasfaser GmbH	Berlin, Germany
<i>Working Student, Strategic Area Development</i>	<i>June 2024 – Present</i>
• Improved operational efficiency by 20% through automated geospatial analysis using QGIS and Python for fiber network deployment. • Conduct cost-benefit analyses and site assessments to identify optimal deployment areas while navigating geographical constraints and market competition. • Developed Python-based geocoding tool for German addresses, reducing processing time by 30%.	
Department of Civil Engineering, RUET	Rajshahi, Bangladesh
<i>Project Assistant (GIS and Geospatial Planner)</i>	<i>February 2020 – April 2020</i>
• Led an airport master planning project using GIS analysis to improve regional connectivity for underserved communities in northwestern Bangladesh. • Collaborated with engineering teams and stakeholders to integrate spatial analysis into transportation planning decisions.	

Research Experience

Flood Vulnerability Assessment Using Geospatial Analysis	<i>November 2018</i>
<i>Bachelor's Thesis: "Application of Remote Sensing and GIS for Vulnerability Assessment: A Case Study of Bogra, Sirajganj, and Tangail Districts"</i>	
• Conducted vulnerability assessment for three flood-prone districts in Bangladesh using satellite imagery to identify at-risk communities. • Developed geospatial methodologies and produced flood risk maps to support disaster planning.	

Personal Projects

Forest Loss Analysis in Colombia's Indigenous Lands
github.com/rafimt/Deforestation-Analysis
• Analyzed 10 years of satellite images using Google Earth Engine and Python to track deforestation and its effects on vulnerable communities, revealing a 15% loss in tree cover.
Wildfire Data Analysis
github.com/rafimt/Wildfire_Analysis_2022
• Developed a Python tool to analyze wildfire patterns in 8 countries (analyzing 1,234 fires total) to help with disaster planning.
Berlin's Environmental Dynamics Analysis
github.com/rafimt/Project-Geoinformatics-2024
• Assessed Urban Heat Island effects using satellite data, finding that built-up areas experienced the highest temperatures compared to other land types.

Skills and Tools

• Core Technical: GIS, Remote Sensing, Spatial Analysis, Geodatabase Maintenance, Spatial Data Management, Field Research & Data Collection
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- **QGIS:** Data Management, Coordinate Reference Systems, Map Creation and Cartography, Spatial Joins, Plugin Usage, Database Connectivity, Processing Modeler, Python Scripting and Workflow Automation
- **Programming:** Python (Pandas, Geopandas, Rasterio, OSMnx, Folium, Scikit-learn, Seaborn, Pytorch, Matplotlib), SQL, JavaScript
- **Tools & Software:** Google Earth Engine, Git, Jupyter Notebook, Excel, PostgreSQL/PostGIS, MATLAB
- **Web GIS:** Django, HTML, CSS, Leaflet, Bootstrap
- **Machine Learning:** Supervised/Unsupervised Learning, Neural Networks, Model Evaluation, Classification, Object Detection, CNN, Vision Transformer, Graph Neural Network
- **Geospatial Expertise:** Engineering Survey, Least Squares Adjustment, Cartographic Principles, GNSS Processing, Environmental Analysis, Vulnerability Assessment

Awards and Recognition

- **Planning and Design of Compact Township, 2nd Runner Up**
Inter-University Undergraduate Planning Students Competition, Bangladesh University of Engineering and Technology (BUET)
- **Publication**
“Identification of Determinant Factors Influencing Modal Choice Behavior and Satisfaction Level of Commuters: A Case of Rajshahi City”
Trends in Civil Engineering and its Architecture, Lupine Publishers
DOI: 10.32474/TCEIA.2019.03.000171

Languages

English (Professional working proficiency), **German** (Elementary proficiency - A2)